

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
To solve this problem , I first need to find out how long it takes Carla to download the file normally .
Since Carla can download 2 GB / minute , I need to divide the total size of the file (200 GB) by the
download speed (2 GB / minute) to get the normal download time .

Normal download time = Total size of the file / Download speed = 200 GB / 2 GB / minute = 100
minutes .

Now , I need to consider the interruption caused by Windows . Carla is 40 % of the way through the
download when the interruption happens . This means she has downloaded 40 % of the file . To find out
how much of the file she has downloaded , I need to calculate 40 % of 200 GB .

40 % of 200 GB = 0 . 4 * 200 GB = 80 GB .

So , Carla has downloaded 80 GB before the interruption . This means she still has 200 GB - 80 GB =
120 GB left to download .

After the interruption , Carla has to start the download from the beginning , which means she will
download the entire 200 GB again .

Now , I need to find out how long it takes Carla to download the remaining 120 GB . Since she can
download 2 GB / minute , I can divide the remaining size of the file (120 GB) by the download speed (2
GB / minute) to get the time it takes to download the remaining part .

Time to download the remaining 120 GB = Remaining size of the file / Download speed = 120 GB / 2 GB
/ minute = 60 minutes .

However , Carla also has to download the first 80 GB again , which took her 40 minutes initially (80 GB
/ 2 GB / minute = 40 minutes) .

So , the total time it takes Carla to download the entire file again is the sum of the time it takes to
download the remaining 120 GB and the time it took to download the first 80 GB .

Total time = Time to download the remaining 120 GB + Time to download the first 80 GB = 60 minutes
+ 40 minutes = 100 minutes .

Since Carla had already downloaded 40 % of the file before the interruption , she effectively has to
download the whole file again , which takes 100 minutes . However , she has already downloaded 40 %
of the file , so she doesn 't need to download that part again .

To find the effective download time , I need to subtract the time it took to download the 40 % (80 GB)
from the total download time .

Effective download time = Total download time - Time to download the 40 % = 100 minutes - 40
minutes = 60 minutes .

</think>< answer> 60 </answer>